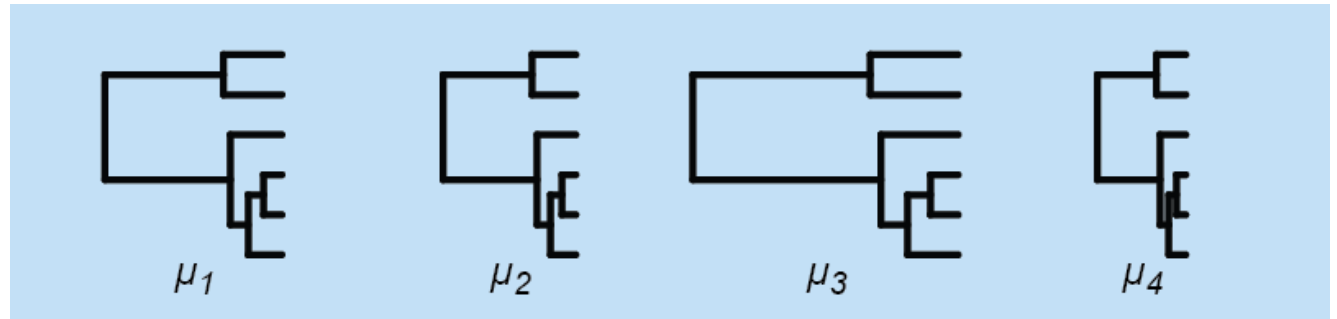

Molecular clock dating and modelling rate variation

Accounting for evolutionary rate variation
in passerine birds

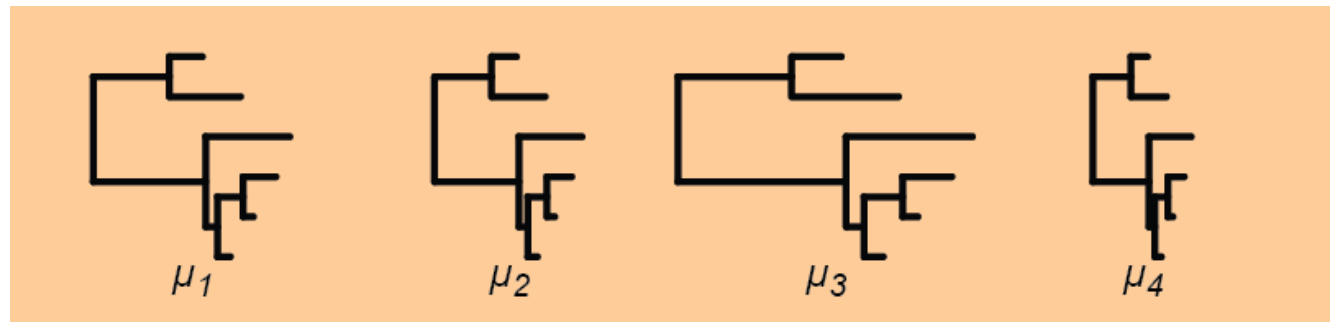
Introduction

Evolutionary rate variation

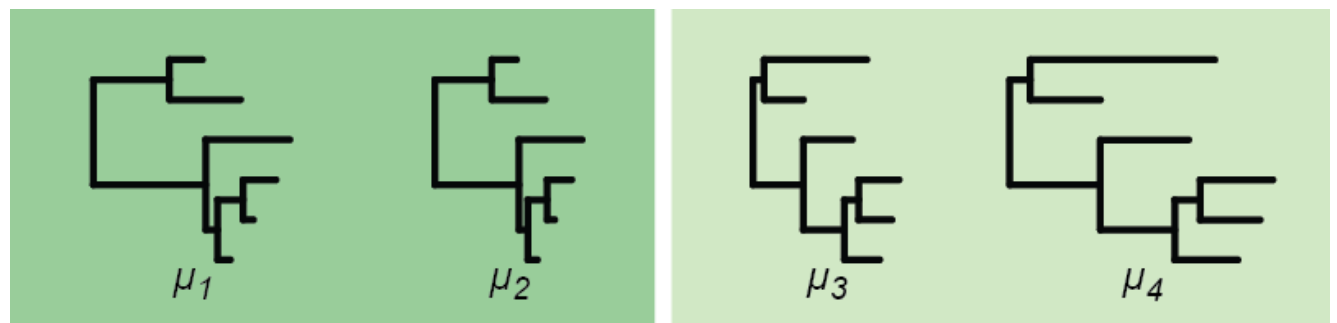
Across loci



Across lineages

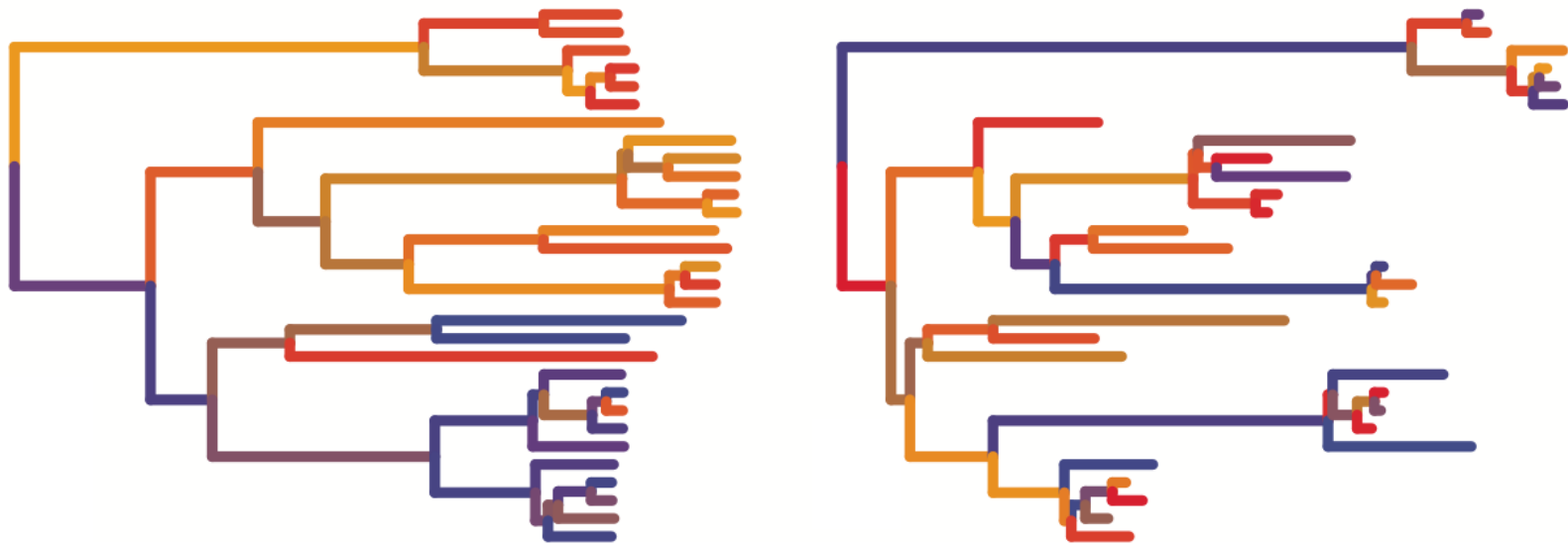


Across loci x
across lineages



Evolutionary rate variation

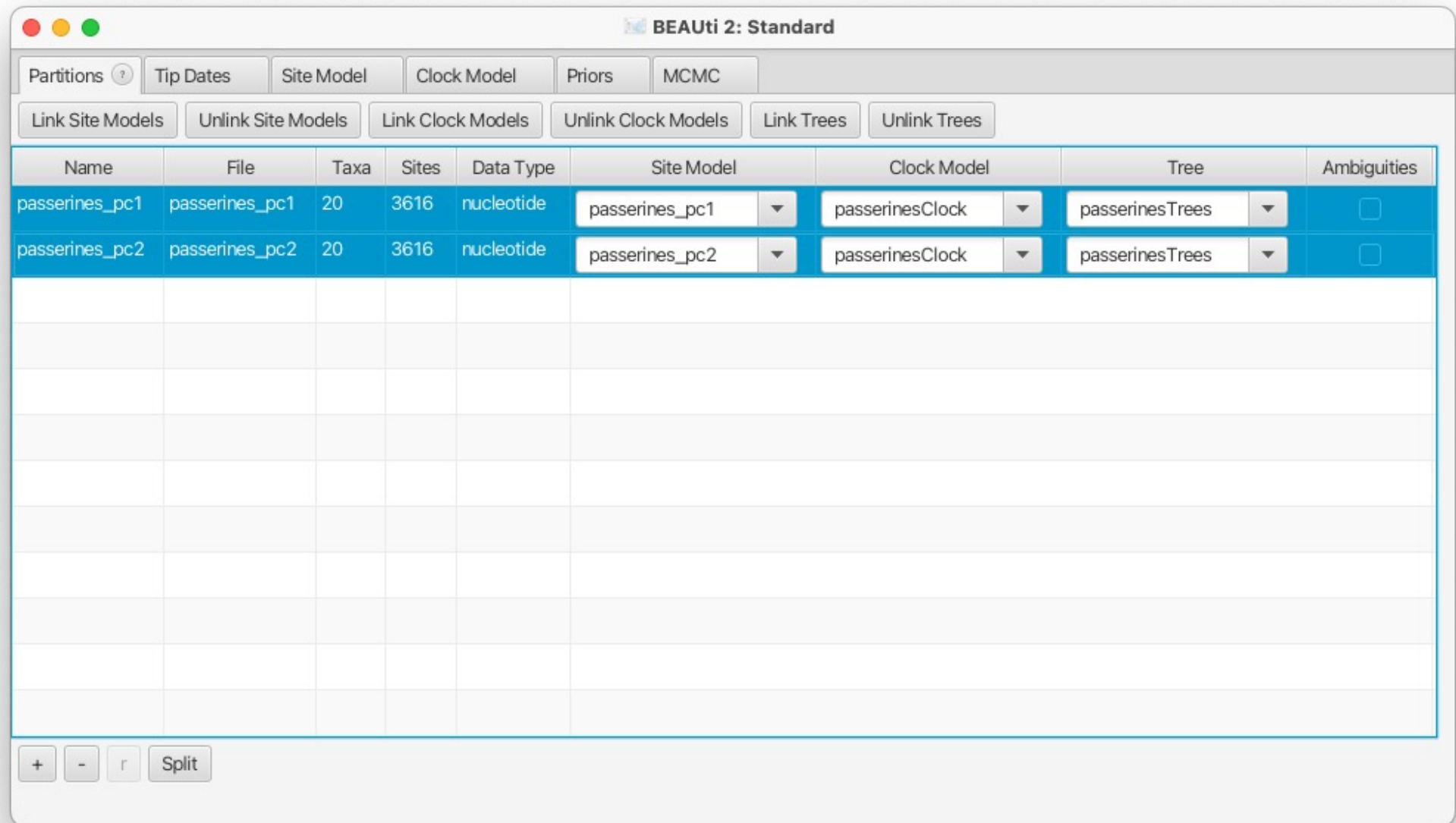
- Bayesian phylogenetic analysis
 - Rate variation across sites
 - Rate variation across loci (or data subsets)
 - Rate variation across branches



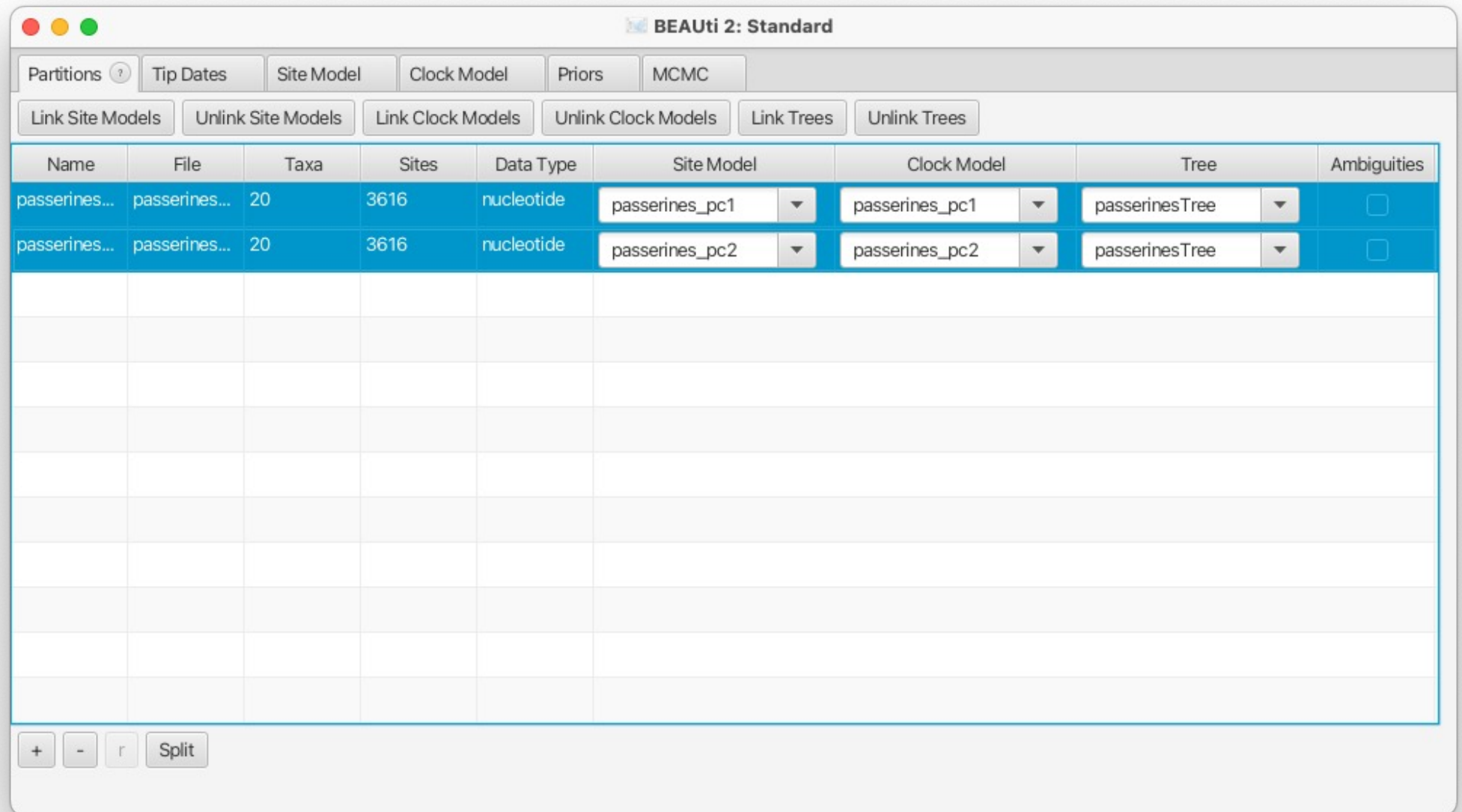
Passerine birds



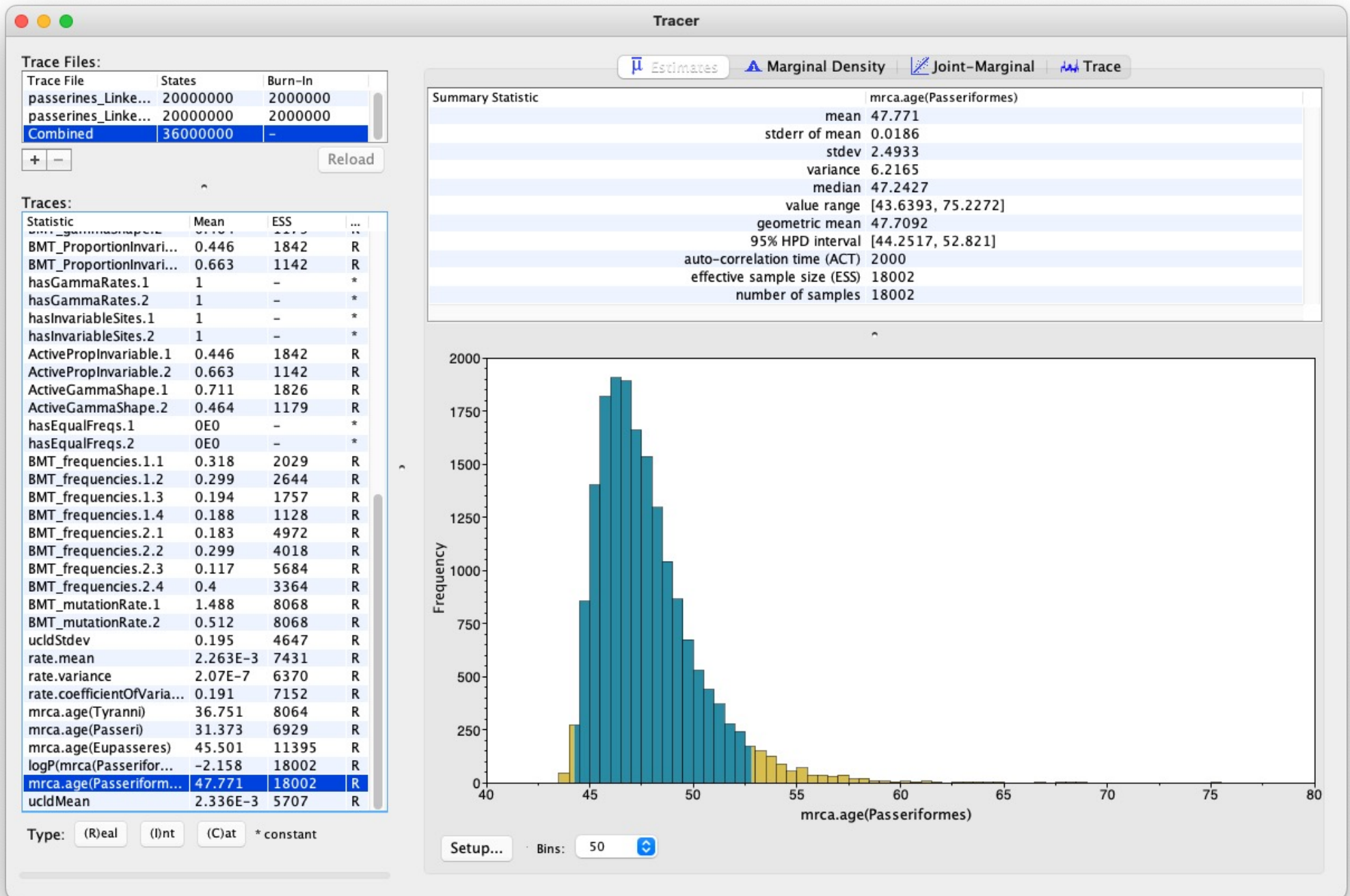
Linked relaxed clocks

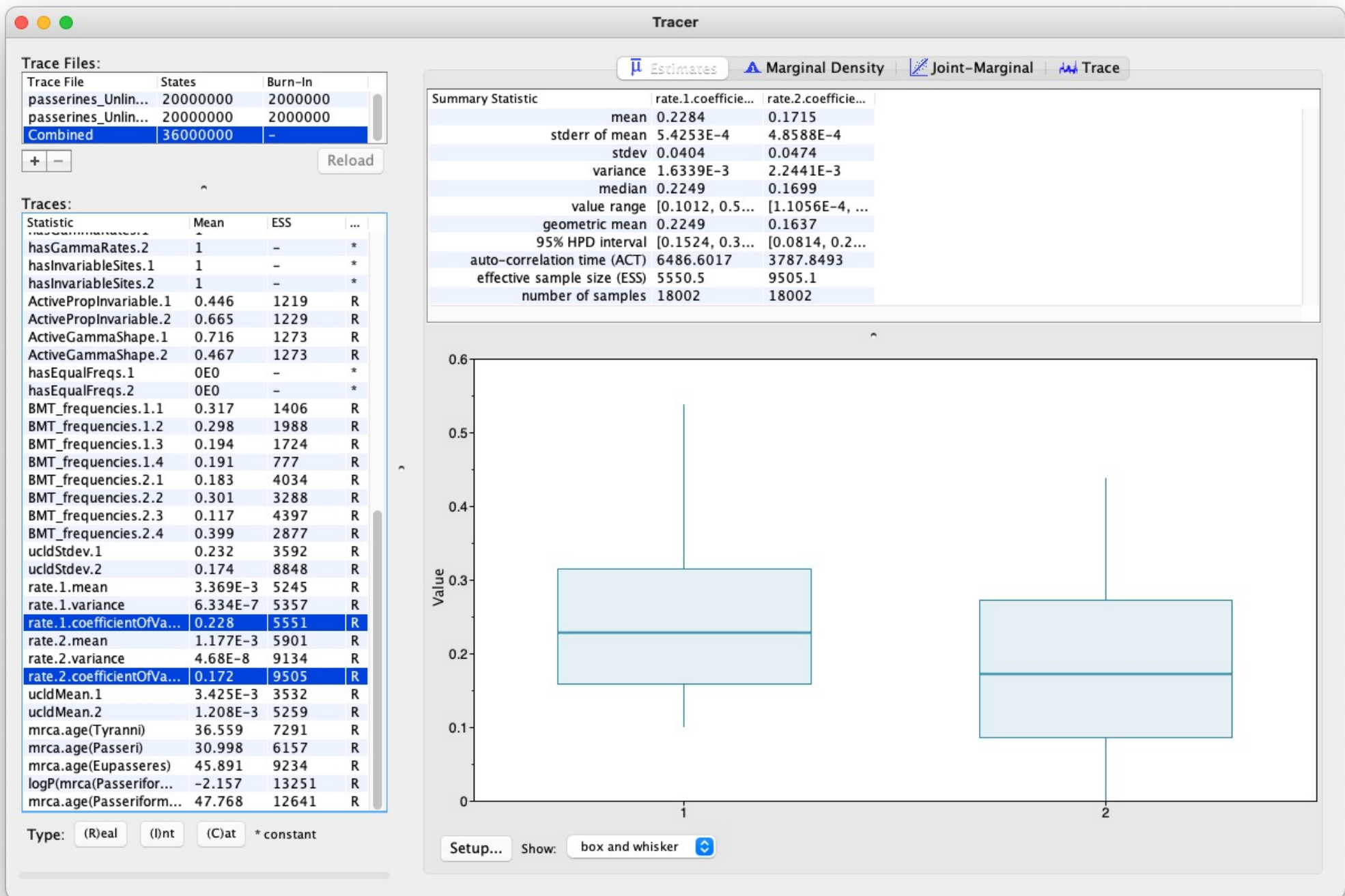


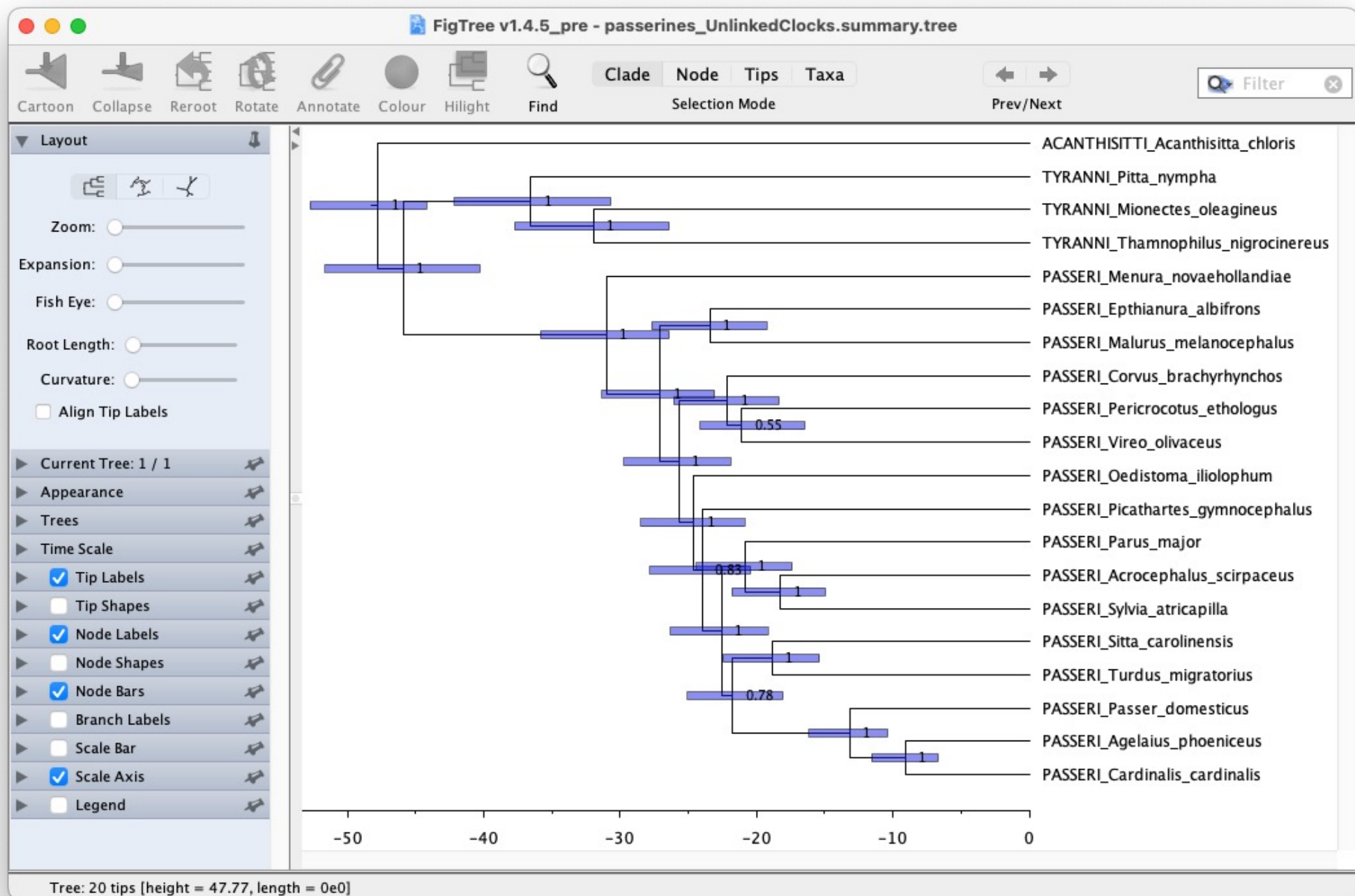
Unlinked relaxed clocks



Summary of outcomes







Outcomes

- Rate variation across nucleotide sites modelled using a gamma distribution
- Rate variation across codon sites modelled using parameters for relative rates (linked clocks) or absolute rates (unlinked clocks)
- Rate variation across branches modelled using uncorrelated lognormal relaxed clocks